

Phase 4 — Dashboard

Final Implementation Report

CareerBuddies Founder Finance · read-only dashboard over the Phase 3 engine
Branch claude/compassionate-bardeen-9zxrjv · code commit d7fcfe6
Final status: **PHASE 4: APPROVED** · Phase 5 not started

1 Project name

CareerBuddies Founder Finance (Founder Ledger) — monorepo `CB-Founder-Ledger/` , repository `d4deepak-2005/career-buddies-website` .

2 Phase 4 scope

The Founder Finance **Dashboard**: a read-only presentation layer over the verified Phase 3 engine (Option C reimbursements unchanged). Not built: approvals (5), recurring (6), reports/exports (7), audit log, hardening (8), launch (9), or any new accounting rule.

3 Product Plan requirements used

Product Plan (\$6, \$16, \$21)	Delivered
KPI cards: Total Investment, Total Business Expenses, Outstanding Settlements, Founder Capital	4 KPI cards + 4 supporting figures
Founder contribution bar chart	Contribution bars (capital vs loans)
Expense-by-category donut/pie	Donut with legend, shares and data table
Monthly expense and investment line chart	Two-series line chart with data table
Founder cards: Invested, Fair Share, Net Position, To Pay/To Receive	Founder cards using engine values
Recent transactions table	Table (≥ md) / card list (mobile)
Settlement summary with direct Settle action	Recommendations + Settle button (prefilled Settlement form)

Product Plan (\$6, \$16, \$21)	Delivered
Date/month filters, founder/category filters	Period presets, custom range, founder, category
All numbers calculated from the database	One server endpoint; no client totals
Charts: Recharts or Chart.js (recommended)	UI decision: hand-drawn accessible SVG/CSS — no new dependency

4 Dashboard architecture

Browser → `GET /api/dashboard` → server loads founders + transactions once → Phase 3 engine runs (as of the period end) → pure `domain/dashboard.ts` selects and sums *engine results* for the period/filters → JSON. React formats and draws; a test forbids money arithmetic and `.reduce` in the dashboard files. Filters live in the URL query string and are validated by the server, so every widget uses the same period.

5 Dashboard components

Component	Role
DashboardPage	Filter bar, KPI cards, panels, founder cards, settlement summary, recent list; loading/error/empty states
charts.tsx	ContributionBars, CategoryDonut, MonthlyLines (pixel geometry only) + screen-reader data tables
periods.ts	Preset date ranges (dates only, no money)
TransactionForm	Reads Settle prefill (type, payer, receiver, amount); values re-validated by the server

6 Backend / API changes

Change	Detail
<code>GET /api/dashboard</code>	Authenticated; query <code>from, to, founderId, categoryId</code> (strict Zod; real dates; <code>from ≤ to</code> ; canonical ObjectIds; unknown founder/category → 404). No write methods
<code>domain/dashboard.ts</code> (new, pure)	Period/filter selection and summing of engine results
<code>loadCalculation({asOf})</code>	Optional: feed the engine transactions dated <code>≤ period end</code> (view parameter, no new rule)
Categories	Selected with the ledger rows; the donut tail is merged server-side into "Other categories"

7 Calculation sources

Every figure comes from the Phase 3 engine (`calculationEngine.ts`): positions, expense breakdowns (Option C founder-funded vs reimbursed), effects (contribution, loan, refund, settlement) and recommendations. There is no second calculation path; the dashboard only filters and sums those results.

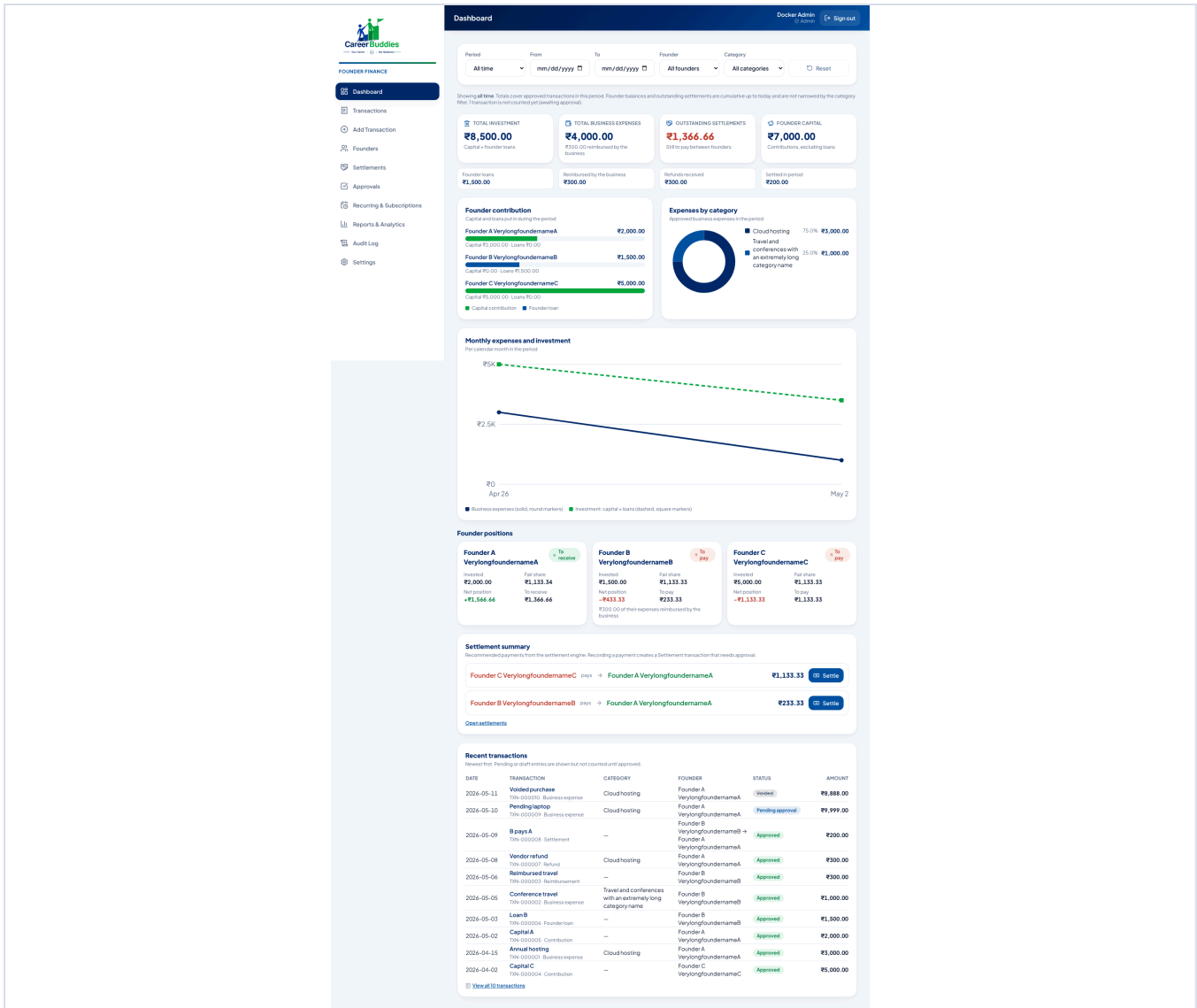
8 Financial KPI definitions

KPI	Definition	Label
Total Business Expenses	Σ gross approved expenses in period (+ "reimbursed by the business", founder-funded = difference)	Implementation assumption
Total Investment	Capital contributions + founder-loan principal in period	Implementation assumption
Founder Capital	Capital contributions only (loans are expected to be repaid, PDF §7)	Implementation assumption
Outstanding Settlements	Engine total payable (= receivable) after settlements, cumulative to period end	Phase 3 value
Reimbursed by the business	Valid linked reimbursements of the period's expenses	Option C / implementation assumption
Refunds, Settled, Loans	Σ of counted refunds / settlements / loans in period	Phase 3 effects
Net position	Paid – Fair share (unchanged)	Inferred from PDF example

The PDF lists the widgets but not these formulas; they are documented in `docs/PHASE-4-DASHBOARD.md` and need product-owner confirmation.

9 Founder balance presentation

Cards show Invested (capital + loans), Fair share, Net position (explicit +/-) and To pay / To receive with a text badge (never colour alone). Values are the engine's, cumulative to the period end, and are not narrowed by the category filter. Verified: Σ net = 0; Option C example A +1,333.33 / B -666.67 / C -666.66.



Desktop — all time (engine-verified figures)

10 Settlement presentation

"Who pays whom" lists the engine's recommendations (never recomputed); each row has a **Settle** button that opens Add Transaction prefilled as a Settlement. It does not record anything by itself; the normal validated form is submitted and the settlement later needs Phase 5 approval. A settlement is never shown as an expense.

11 Reimbursement presentation

Approved Option C wording only: "reimbursed by the business" on the expense KPI hint, the supporting figure and per-founder notes. No `external`, `IA-5` or `PASS_WITH_EXTERNAL` text or field anywhere (asserted in server, client and browser tests).

12 Transaction activity

Newest 10 matching transactions (any status) with date, description, number, type, category, payer → payee, status badge and amount; pending/draft/voided rows are shown but flagged as not counted. A link opens the full Transactions module.

13 Filters

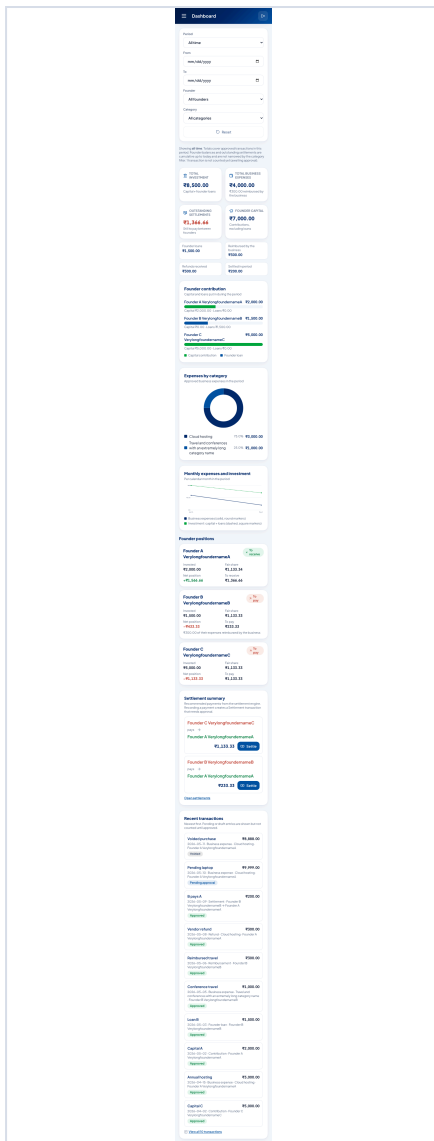
Period presets (All time, This month, Last month, Last 3 months, This year) or custom From/To; founder; category; Reset. Sent to the server; **balances** are cumulative up to the period end, **period figures** cover the inclusive range. A scope note under the filters states this on every view. Founder filter = payer or counterparty; category filter counts only that category. Server verified: inclusive bounds, as-of balances, founder/category narrowing, invalid values rejected.

14 Security

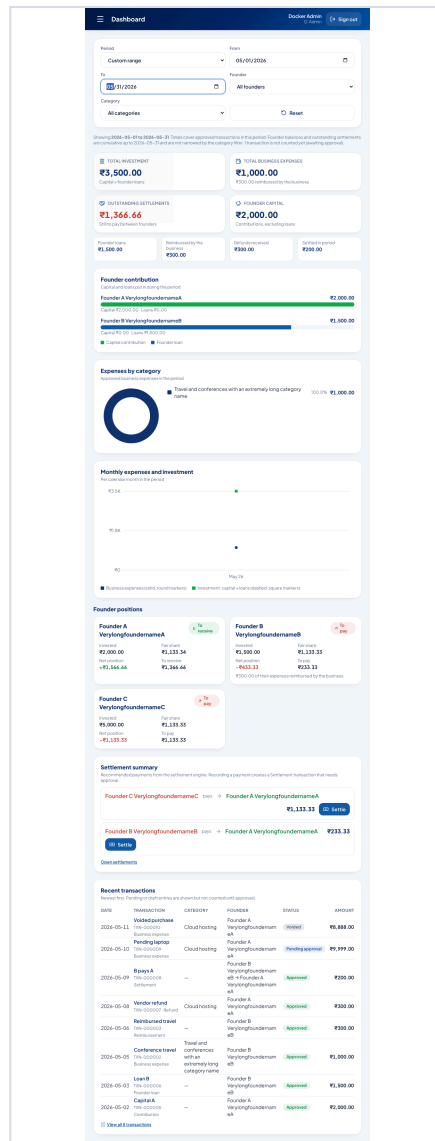
- Authentication required (401 otherwise); founders and admins can read (same visibility as the ledger); no write method exists.
- Strict query schema: client-supplied totals (`totalBusinessExpensesMinor` , `kpis` , `netPositionMinor` ...), unknown keys, repeated keys, `$ne` operators, malformed dates/ids all return 400; unknown founder/category 404.
- No financial value is trusted from the client; responses contain no secrets or internal errors.
- All earlier controls unchanged (cookies, rate limits, RBAC, `sanitizeFilter`). No security setting was weakened for testing.

15 Responsive design

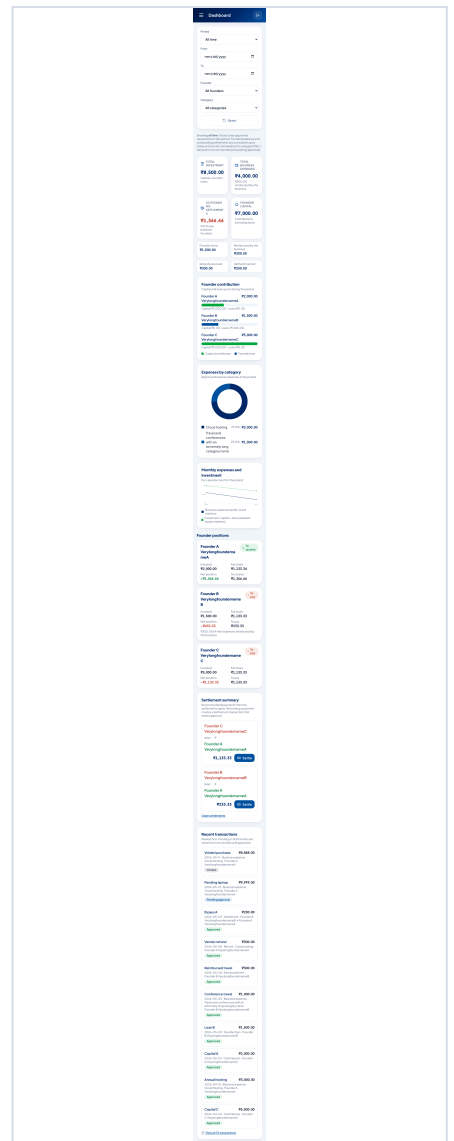
Cards stack 4→2→1 columns; founder cards 3→2→1; table becomes a card list below 768px; long founder and category names wrap (`break-words` , `min-w-0`). Tested at 1366×900, 820×1180, 390×844 and 320×640. **Real defect found and fixed by browser testing:** the visually hidden chart data tables were absolutely-positioned bare `<table>` elements that kept their content width and widened the page (499px on a 390px screen); wrapping them in a clipped `sr-only` div fixed it (verified `scrollWidth` = viewport at all sizes). Earlier Founders-page overflow fix remains.



Mobile 390 px



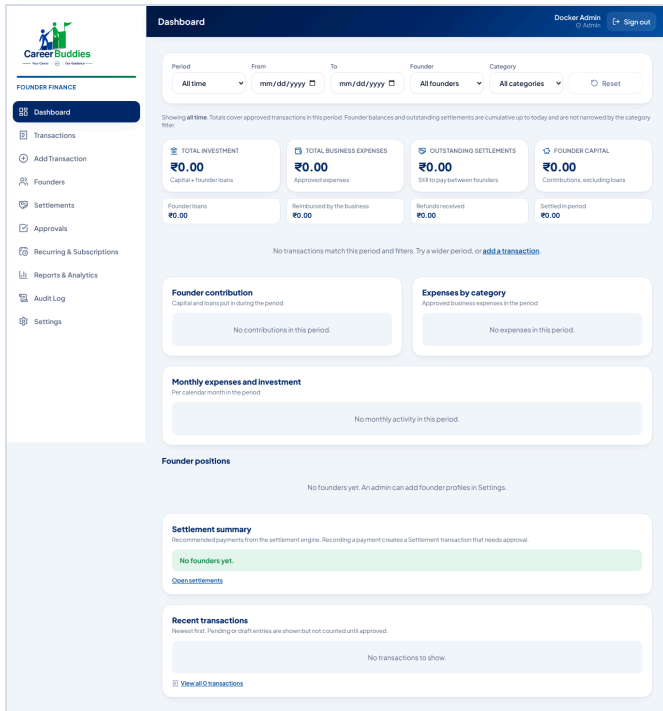
Tablet 820 px — May period



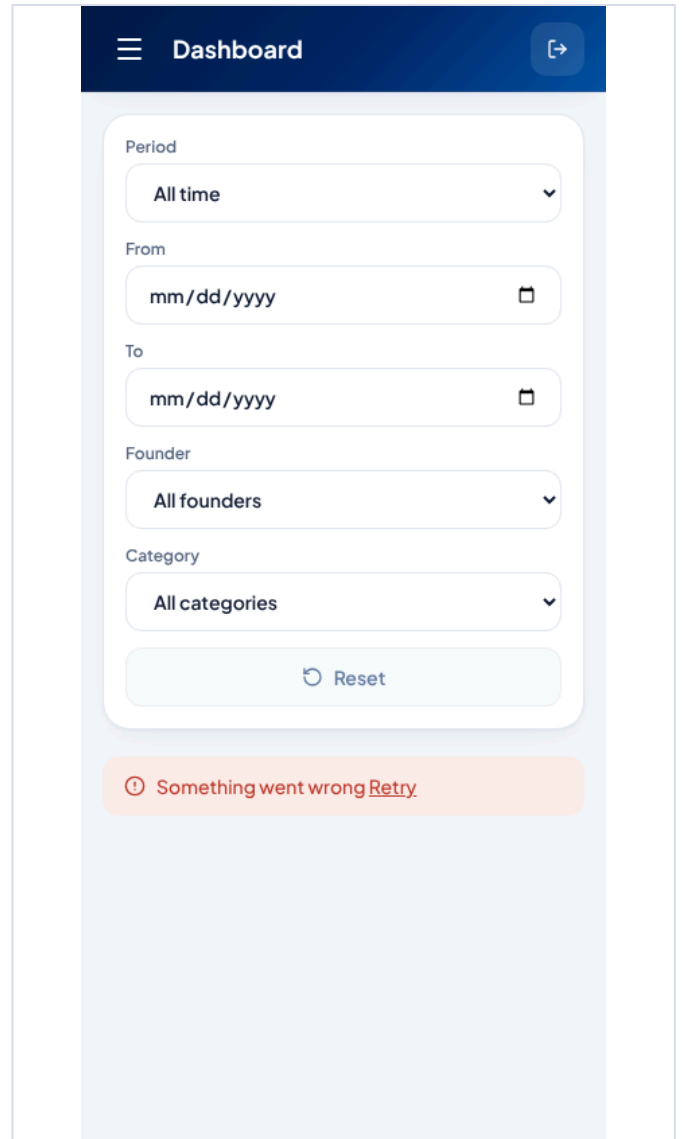
Narrow 320 px

16 Empty, loading and error states

Loading: skeleton (`role=status`). Error: friendly message, retry, no stack trace; filters stay usable; the founder/category lists load independently so their failure never blanks the dashboard. Empty: per-section messages ("No contributions in this period", "No founders yet", "No payments needed..."), no chart drawn without data; ₹0.00 shown only when the server returns zero.



Empty database



API error, mobile

17 Exact test counts

Suite	Result
Server	298 / 298 (18 files; +19 dashboard)
Client	67 / 67 (6 files; +12 dashboard, +2 settle prefill)
Typecheck / production build	clean / OK
Docker smoke	73 / 73 (12 new dashboard checks)
Docker reimbursement E2E (regression)	22 / 22
Browser (dashboard)	144 / 144 — desktop, tablet, mobile, narrow (run on a fresh stack)
Responsive stress	200 page loads, 0 overflows

18 Server tests

298 passed / 298. `dashboard.test.ts` (19): auth 401, founder/admin read, read-only, 15 invalid-query variants, 404s, zero data, empty period, expense/contribution/loan/reimbursement/refund/settlement

totals, founder cards equal the Phase 3 endpoint, Σ net = 0, recommendations equal the settlements endpoint, Option C figures (+133,333 / -66,667 / -66,666), voided/pending/draft/rejected/Other excluded, inclusive period bounds, as-of balances, founder and category filters, recent ordering/limit, uncategorised label, donut grouping and 9-billion values with month gaps. All Phase 1-3 suites (engine 50, accounting review 37, reimbursements 25, invariants 8, financials 19, transactions 40, rbac 22 ...) pass unchanged.

19 Client tests

67 passed / 67. Dashboard (12): server values shown verbatim with deliberately inconsistent data (no client arithmetic), Option C wording and no "external", attention banner, loading, error + retry, empty states, failing filter lists, filters sent as query parameters and Reset, server 400 shown, period helpers incl. leap years/year boundaries, no-arithmetic guard. Transactions: Settle prefill and malformed-prefill ignored. Phase 1-3 client suites pass unchanged.

20 API tests

The 19 dashboard API tests run through HTTP on MongoDB 7 (supertest), plus the 73-check Docker smoke over real HTTP against the containers (dashboard 401/200, valid filters, bad dates, reversed range, client totals, operator injection, unknown founder, no write method, pending not counted, no obsolete fields).

21 MongoDB tests

Server tests use a real MongoDB 7 container. Direct inspection of the Docker MongoDB after the browser run: 10 transactions (8 approved, 1 pending, 1 voided); Σ approved expense amounts **400,000** = dashboard KPI 400,000 (independent database sum); no reimbursement without a link; reservation counters match ("*No drift*"). The dev database holds no test data; stacks and volumes were removed.

22 Docker tests

Images built; Compose healthy (client, mongo, server); health reports the database connected. Stack 1 (fresh): smoke 73/73 and the Option C reimbursement E2E 22/22. Stack 2 (fresh): dashboard browser suite 144/144 against real MongoDB data; authenticated dashboard API and rendering verified through nginx at `http://localhost:8080`.

23 Browser tests

Real Chromium (Playwright), fresh stack: empty state first (zero KPIs, "No founders yet", no fake chart); then fixtures → hand-computed expectations → API → rendered text, all equal, per viewport: menu navigation to Dashboard, KPI cards, founder cards (fair share, net sign, receive/pay), settlement rows with exact amounts, recent list (pending flagged), charts with data tables, May period filter (expenses 1,000.00; investment 3,500.00; outstanding unchanged 1,366.66), founder filter, Reset, empty period, loading state (delayed API), error state (500) and retry, Settle prefilled to 1133.33, rendered KPIs equal the API, no console errors, no obsolete wording. Sample hand-computed result: A paid 2,700 / fair 1,133.34 / net +1,566.66; B net -433.33; C net -1,133.33; after B's 200 settlement C→A 1,133.33 and B→A 233.33.

24 Responsive tests

Desktop 1366×900, tablet 820×1180, mobile 390×844, narrow 320×640: no horizontal overflow on the dashboard (all-time, May, empty period, error state) and the Settle form — plus a stress run of 200 page loads (dashboard, dashboard with filters, founders, settlements, add transaction × 10 × 4 sizes): 0 overflows.

25 Public / local testing URL

PUBLIC DEPLOYMENT STATUS: Not possible. The sandbox has policy-proxied outbound access only (probe to a tunnel service returned HTTP 403/000), no inbound exposure and no tunnel tooling. No public deployment was attempted and no public URL is claimed.

LOCAL TESTING URL: `http://localhost:8080` (verified in a real browser; health `/api/health`). Recreate:

```
cd CB-Founder-Ledger
cp .env.example .env # set JWT_ACCESS_SECRET, MONGO_ROOT_PASSWORD, SEED_ADMIN_EMAIL,
SEED_ADMIN_PASSWORD; plain HTTP: NODE_ENV=development, COOKIE_SECURE=false
docker compose --env-file .env -f docker/docker-compose.yml up --build -d
docker compose --env-file .env -f docker/docker-compose.yml exec server node
dist/scripts/seedAdmin.js
# optional demo/verification data: SMOKE_ADMIN_EMAIL=... SMOKE_ADMIN_PASSWORD=... node
tests/browser/dashboard-browser.mjs (fresh stack)
```

Until Phase 5 nothing can be approved through the app, so a real deployment shows zeros; the browser suite approves its fixtures with a direct database write.

26 Test credentials / method

Disposable local-only admin used for verification (stack and volume destroyed afterwards):

`admin@careerbuddies.test` / `Docker-verify-passphrase-9`; valid only on a stack seeded with those values. No production credentials, secrets, API keys or database credentials are in this report or the repository (`.env` is git-ignored). MongoDB is never published.

27 Screenshots

Included above (desktop, tablet, mobile, narrow, empty, error). Rendered from the real Docker stack during the final browser run.

28 Files changed

21 files, +1,347 / -13 (code commit `d7fcfe6`).

Area	Files
Server	domain/dashboard.ts (new), modules/dashboard/dashboard.routes.ts (new), modules/financials/financials.service.ts (asOf, categoryId), app.ts, tests/dashboard.test.ts (new)
Client	features/dashboard/{DashboardPage, charts, periods}.ts(x) + dashboard.test.tsx (new), App.tsx, TransactionForm.tsx (Settle prefill), lib/{types, money}.ts, transactions.test.tsx
Verification	tests/browser/dashboard-browser.mjs (new), tests/smoke-docker.sh

Area	Files
Docs	README.md, docs/{ARCHITECTURE, PHASE-3-CALCULATION-SPEC, PHASE-3-OPEN-DECISIONS}.md, docs/PHASE-4-DASHBOARD.md (new), this report

29 Documentation updated

New `docs/PHASE-4-DASHBOARD.md` (requirements, API, every definition labelled Product Plan / Phase 3 rule / implementation assumption / UI decision / limitation, worked reconciliation). Updated README, ARCHITECTURE, PHASE-3-CALCULATION-SPEC (§10 consumers), PHASE-3-OPEN-DECISIONS (Phase 4 note). Approved Option C rules were not changed.

30 Assumptions

- **Implementation assumptions (need product-owner confirmation):** Total Investment = capital + loans; Founder Capital = capital only; balances cumulative to period end vs period flows in [from, to]; reimbursed amount attributed to the expense's period; "Invested" on founder cards = capital + loans.
- **UI decisions:** SVG charts instead of Recharts/Chart.js; founder filter = payer or counterparty; category filter counts only that category; donut limited to 7 slices; Settle prefills (does not record).
- **Unchanged Phase 3 rules:** approved-only, Option C, Net = Paid – Fair share (inferred from PDF example).

31 Limitations

- Nothing can be approved in the app until Phase 5, so real deployments show zeros; tests approve via DB writes.
- One request feeds all widgets: if it fails the figure area shows an error (filter lists are independent).
- Charts have no hover tooltips (values are printed and available as tables).
- No public deployment (environment).

32 Known issues

- No public URL.
- The in-memory login rate limiter (10 / 15 min / IP) means repeated automated runs need a server restart; the browser suite logs in once and reuses the session.
- The Docker smoke and E2E scripts expect an empty database and the dashboard browser suite expects a fresh stack; run each on a fresh stack (the verification did).
- No other defects known. Two real responsive defects were found by the browser tests and fixed (chart data tables; earlier long names).

33 Git branch

`claude/compassionate-bardeen-9zxrjv`

34 Git commit

Dashboard code and tests: `d7fcfe6`. This PDF is committed immediately afterwards (a commit cannot contain its own hash); its hash is reported in the final message. Working tree clean after push.

35 Final status

PHASE 4: APPROVED

All required verification passed (server 298/298, client 67/67, smoke 73/73, E2E 22/22, browser 144/144, stress 0 overflows, MongoDB inspected). The only gap is the unavailable public URL (environment limitation).

36 Phase 5 status

Phase 5 was NOT started. No approval workflow, recurring/subscriptions, reports/exports, audit-log or later-phase work was done.